Time-Series and Cross-Sectional Stock Return Forecasting: New Machine Learning Methods

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Abstract

This paper extends the machine learning methods developed in Han et al. (2019) for forecasting cross-sectional stock returns to a time-series context. The methods use the elastic net to refine the simple combination return forecast from Rapach et al. (2010). In a time-series application focused on forecasting the US market excess return using a large number of potential predictors, we find that the elastic net refinement substantively improves the simple combination forecast, thereby providing one of the best market excess return forecasts to date. We also discuss the cross-sectional return forecasts developed in Han et al. (2019), highlighting how machine learning methods can be used to improve combination forecasts in both the time-series and cross-sectional dimensions. Overall, because many important questions in finance are related to time-series or cross-sectional return forecasts, the machine learning methods discussed in this paper should provide valuable tools to researchers and practitioners alike.

JEL classifications: C53, G11, G12

Key words: Expected stock returns, Time series, Cross section, Forecast combination, Shrinkage, Elastic net

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1 Introduction

Researchers in finance increasingly rely on machine learning techniques to analyze big data. The initial application of the *least absolute shrinkage and selection operator* (Tibshirani 1996, LASSO)—one of the most popular machine learning techniques—in finance appears to be Rapach et al. (2013), who analyze lead-lag relationships among monthly international equity returns in a high-dimensional setting. More recently, Gu et al. (2018) employ a comprehensive set of machine learning tools, including the LASSO, to analyze the time-series predictability of monthly individual stock returns, while Chinco et al. (2019) use the LASSO to predict individual stock returns one-minute ahead. Freyberger et al. (forthcoming) apply a nonparametric version of the LASSO to accommodate nonlinear relationships between numerous firm characteristics and cross-sectional stock returns. Kozak et al. (forthcoming) utilize the LASSO in a Bayesian context to model the stochastic discount factor using a large number of firm characteristics. Incorporating insights from Bates and Granger (1969) and Diebold and Shin (forthcoming), Han et al. (2019) propose procedures for forecasting cross-sectional returns using the information in more than 100 firm characteristics.[1]

In this paper, we show how the approach of Han et al. (2019), originally designed for forecasting cross-sectional stock returns, can be modified for time-series forecasting of the market excess return. A voluminous literature investigates market excess return predictability based on a wide variety of predictor variables.[2] In the presence of a large number of potential predictor variables, conventional forecasting methods are susceptible to in-sample overfitting, which often translates into poor out-of-sample performance. Rapach et al. (2010) employ forecast combination (Bates and Granger 1969) to incorporate the information in a large number of predictor variables in a manner than guards against overfitting. They find that a simple combination forecast—the average of univariate predictive regression forecasts

[1]We focus on numerical data in this paper. For textual analysis, see, e.g., Tetlock (2007), Loughran and McDonald (2011), and Ke et al. (2019).

[2]See Rapach and Zhou (2013) for an extensive survey of the literature.

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based on the individual predictor variables—substantially improves out-of-sample forecasts of the US market excess return. Extending the methods of Han et al. (2019) to a time-series context, we describe how the elastic net (Zou and Hastie 2005, ENet), a well-known variant of the LASSO, can be used to refine the simple combination forecast, resulting in what we call the combination ENet (C-ENet) forecast. Intuitively, as explained by Han et al. (2019), the ENet refinement allows us to more efficiently use the information in the predictor variables by selecting the most relevant predictors to include in the combination forecast. In an empirical application, we show that the C-ENet approach indeed improves the accuracy of US market excess return forecasts and provides substantive economic value to a mean-variance investor. Overall, our C-ENet forecast appears to be among the best market excess return forecasts to date.

The rest of the paper is organized as follows. Section 2 describes the construction of market excess return forecasts, including the C-ENet forecast. Section 3 reports results for an empirical application centered on forecasting the US market excess return using a variety of predictor variables from the literature. Section 4 outlines the construction of the cross-sectional return forecasts proposed by Han et al. (2019). Section 5 concludes.

2 Time-Series Return Forecasts

2.1 Predictive Regression

Stock market excess return predictability is typically analyzed in the context of a univariate predictive regression model:

$$r_t = \alpha + \beta x_{j,t-1} + \varepsilon_t, \tag{2.1}$$

where r_t is the period-t return on a broad stock market index in excess of the risk-free return, $x_{j,t}$ is a predictor variable, and ε_t is a zero-mean disturbance term. It is straightforward to use Equation (2.1) to generate an out-of-sample forecast of r_{t+1} based on $x_{j,t}$ and data available

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through period t:

$$\hat{r}_{t+1|t}^{(j)} = \hat{\alpha}_{1:t}^{(j)} + \hat{\beta}_{1:t}^{(j)} x_{j,t}, \tag{2.2}$$

where $\hat{\alpha}_{1:t}^{(j)}$ and $\hat{\beta}_{1:t}^{(j)}$ are the OLS estimates of α and β, respectively, in Equation (2.1) based on data available from the start of the sample through t (i.e., the period of forecast formation).

Because there are a plethora of plausible predictor variables, it is advisable to aggregate information when forecasting the market excess return. The most obvious approach for incorporating information from multiple predictor variables is to specify a multiple predictive regression model:

$$r_t = \alpha + \sum_{j=1}^{J} \beta_j x_{j,t-1} + \varepsilon_t. \tag{2.3}$$

It is again straightforward use Equation (2.3) to generate an out-of-sample forecast of r_{t+1} based on $x_{j,t}$ for $j = 1, \ldots, J$ and data available through t:

$$\hat{r}_{t+1|t}^{\text{OLS}} = \hat{\alpha}_{1:t}^{\text{OLS}} + \sum_{j=1}^{J} \hat{\beta}_{j,1:t}^{\text{OLS}} x_{j,t}, \tag{2.4}$$

where $\hat{\alpha}_{1:t}^{\text{OLS}}$ and $\hat{\beta}_{j,1:t}^{\text{OLS}}$ are the OLS estimates of α and β_j, respectively, for $j = 1, \ldots, J$ in Equation (2.3) based on data available through t.

Although the out-of-sample market excess return forecasts in Equations (2.2) and (2.4) are easy to obtain, Goyal and Welch (2008) find that such forecasts based on numerous popular predictor variables from the literature fail to outperform the naive prevailing mean benchmark forecast on a consistent basis over time (as judged by the out-of-sample R^2 statistic, which we define below). The prevailing mean forecast ignores information in any predictor variable; it is simply the historical average excess return based on data available through t:

$$\hat{r}_{t+1|t}^{\text{PM}} = \frac{1}{t} \sum_{s=1}^{t} r_s. \tag{2.5}$$

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The prevailing mean forecast corresponds to the following simple data-generating process for the market excess return:

$$r_t = \alpha + \varepsilon_t, \tag{2.6}$$

namely, the constant expected excess return (or random walk with drift) model. The Goyal and Welch (2008) findings pose important challenges for out-of-sample return predictability, as they indicate that exploiting the information in popular predictor variables via conventional regression methods does not improve forecast accuracy.

2.2 Forecast Combination

The study of Goyal and Welch (2008) was influential in stimulating thinking on how to better use the information in predictor variables to forecast the market excess return. With respect to the univariate predictive regression forecast in Equation (2.2), Rapach et al. (2010) argue that it is risky to rely on a single predictor variable, due to factors such as investor learning and structural change. Building on the seminal work of Bates and Granger (1969), Rapach et al. (2010) recommend forecast combination as a strategy for incorporating information from a variety of predictor variables. Forecast combination reduces forecast "risk" by diversifying across individual forecasts, similarly to diversifying across assets to reduce portfolio risk (Timmermann 2006). Specifically, Rapach et al. (2010) consider a combination forecast that takes the form of a simple average of the univariate predictive regression forecasts based on $x_{j,t}$ for $j = 1, \ldots, J$ in Equation (2.2):

$$\hat{r}^{\mathrm{C}}_{t+1|t} = \frac{1}{J} \sum_{j=1}^{J} \hat{r}^{(j)}_{t+1|t}. \tag{2.7}$$

They show that, in contrast to the conventional univariate and multiple predictive regression forecasts in Equations (2.2) and (2.4), respectively, the combination forecast in Equation (2.7) is able to deliver out-of-sample accuracy gains relative to the prevailing mean forecast on a much more consistent basis over time.

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How is it that—unlike the conventional multiple predictive regression forecast in Equation (2.4), which also includes information from $x_{j,t}$ for $j = 1, \ldots, J$—the combination forecast in Equation (2.7) is able to improve out-of-sample performance? Rapach et al. (2010) point out that forecast combination is effectively a strong shrinkage estimator. They show that the combination forecast in Equation (2.7) makes two adjustments to the conventional multiple predictive regression forecast in Equation (2.4): first, it replaces the OLS multiple regression coefficient estimates with their univariate counterparts, which reduces the role of multicollinearity in producing imprecise parameter estimates; second, the combination forecast shrinks the univariate slope coefficients toward zero by the factor $1/J$, thereby shrinking the forecast to the prevailing mean benchmark.

The usefulness of shrinkage for improving out-of-sample market excess return forecasts stems from a delicate balance required for stock return forecasting. On the one hand, we want to incorporate information from a wide variety of potentially relevant predictor variables, especially since we do not want to neglect relevant information and cannot know a priori which predictors are the most relevant. On the other hand, incorporating information from numerous predictors via the multiple prediction regression forecast in Equation (2.4) is inadvisable. Equation (2.4) is based on conventional estimation of the multiple predictive regression model in Equation (2.3), which is susceptible to overfitting. Conventional OLS estimation maximizes the explanatory ability of the model over the estimation sample, which often leads to poor out-of-sample performance. Overfitting concerns are exacerbated as the number of explanatory variables increases and the signal-to-noise ratio in the data decreases. We encounter both of these challenges when forecasting stock returns: there are numerous plausible predictor variables, and the predictable component in returns is inherently limited. The combination forecast in Equation (2.7) apparently provides an effective shrinkage strategy for incorporating information from numerous plausible predictor variables in a manner that avoids overfitting.

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2.3 Elastic Net

Machine learning techniques also provide a means for implementing shrinkage. Indeed, the popular LASSO estimator is a penalized regression approach that is explicitly designed to prevent overfitting via shrinkage. To compute a forecast based on the multiple predictive regression model in Equation (2.3), instead of the OLS objective function, we estimate the coefficients using the LASSO objective function:

$$\underset{\alpha,\beta_1,\ldots,\beta_J\in\mathbb{R}}{\arg\min} \left[\frac{1}{t}\sum_{s=1}^{t}\left(r_s - \alpha - \sum_{j=1}^{J}\beta_j x_{j,s-1} \right)^2 + \lambda \sum_{j=1}^{J}|\beta_j| \right], \tag{2.8}$$

where $\lambda \geq 0$ is a regularization parameter that controls the degree of shrinkage.[3] The first component of the LASSO objective function is the familiar sum of squared fitted residuals, so that the LASSO and OLS estimators coincide when $\lambda = 0$. The regularization parameter λ shrinks the coefficients toward zero. Unlike rigde regression (Hoerl and Kennard 1970), which relies on an ℓ_2 penalty term, the LASSO employs an ℓ_1 penalty, so that it permits shrinkage to exactly zero (for sufficiently large λ). Shrinkage to zero means that the LASSO also performs variable selection, which facilitates the interpretation of the fitted model.

To implement LASSO estimation, it is necessary to choose the value for λ. The most popular approach is K-fold cross validation. However, the selection of the number of folds K and the construction of the folds are largely arbitrary. The Hurvich and Tsai (1989) corrected version of the Akaike information criterion (Akaike 1973, AIC) provides an alternative to K-fold cross validation for choosing λ. The corrected AIC is simpler to use, in that it does not require arbitrary choices for the number and type of folds. Furthermore, Flynn et al. (2013) show that the corrected AIC has good asymptotic and finite-sample properties for choosing λ.

[3] Following standard practice, the predictor variables are standardized to have zero mean and unit variance before entering Equation (2.8). The final parameter estimates reflect the original scales of the predictor variables.

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Zou and Hastie (2005) propose the ENet as a refinement to the LASSO that includes both ℓ_1 and ℓ_2 components in the penalty term. The ENet estimator is defined by the following objective function:

$$\underset{\alpha,\beta_1,\ldots,\beta_J\in\mathbb{R}}{\arg\min}\left\{\frac{1}{t}\sum_{s=1}^{t}\left(r_s-\alpha-\sum_{j=1}^{J}\beta_j x_{j,s-1}\right)^2+\lambda\left[0.5(1-\delta)\sum_{j=1}^{J}\beta_j^2+\delta\sum_{j=1}^{J}|\beta_j|\right]\right\}, \tag{2.9}$$

where $0\leq\delta\leq 1$ is a parameter for blending the ℓ_1 and ℓ_2 components. A potential drawback to the LASSO is that it tends to somewhat arbitrarily select one predictor from a group of highly correlated predictors. In contrast, using $\delta=0.5$ in Equation (2.9) results in a stronger tendency to select the highly correlated predictors as a group (Hastie and Qian 2016). The corrected AIC can again be used to choose λ in Equation (2.9).

A market excess return forecast based on ENet estimation of the multiple predictive regression model in Equation (2.3) is given by

$$\hat{r}_{t+1|t}^{\text{ENet}}=\hat{\alpha}_{1:t}^{\text{ENet}}+\sum_{j=1}^{J}\beta_{j,1:t}^{\text{ENet}}x_{j,t}, \tag{2.10}$$

where $\hat{\alpha}_{1:t}^{\text{ENet}}$ and $\hat{\beta}_{j,1:t}^{\text{ENet}}$ are the ENet estimates of α and β_j, respectively, for $j=1,\ldots,J$ in Equation (2.3). Intuitively, we rely on the shrinkage properties of the ENet to generate a market excess return forecast that incorporates information from a potentially large number of predictor variables in a manner that guards against overfitting. Whether the ENet is an effective shrinkage strategy for forecasting the market excess return is ultimately an empirical issue. We investigate this issue in our empirical application in Section 3.[4]

2.4 Combination Elastic Net

Incorporating insights from Diebold and Shin (forthcoming), we can also use machine learning techniques to refine the combination forecast in Equation (2.7). A potential drawback

[4]We focus on the results for the ENet in our empirical application in Section 3, although the results are qualitatively similar for the LASSO.

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to Equation (2.7) is that it may "overshrink" the forecast to the prevailing mean, thereby neglecting substantive relevant information in the predictor variables. In an effort to improve the combination forecast by exploiting more of the relevant information in the predictor variables (while still avoiding overfitting), we consider the following Granger and Ramanathan (1984) regression:

$$r_t = \eta + \sum_{j=1}^{J} \theta_j \hat{r}^{(j)}_{t|t-1} + \varepsilon_t, \tag{2.11}$$

which we estimate via the elastic net to select the most relevant univariate forecasts to include in the combination forecast.[5] Specifically, to construct the C-ENet forecast, we first need to define an initial in-sample estimation period and corresponding holdout out-of-sample period; let t_1 denote the size of the initial in-sample period. We then proceed in three steps:

Step 1 For each predictor variable, we compute recursive univariate predictive regression forecasts based on Equation (2.2) over the holdout out-of-sample period:

$$\hat{r}^{(j)}_{s|s-1} = \hat{\alpha}^{(j)}_{1:s-1} + \hat{\beta}^{(j)}_{1:s-1} x_{j,s-1}, \tag{2.12}$$

for $s = t_1 + 1, \ldots, t$ and $j = 1, \ldots, J$.

Step 2 We estimate the Granger and Ramanathan (1984) regression in Equation (2.11) via the ENet over the holdout out-of-sample period:

$$r_s = \eta + \sum_{j=1}^{J} \theta_j \hat{r}^{(j)}_{s|s-1} + \varepsilon_s, \tag{2.13}$$

for $s = t_1 + 1, \ldots, t$. Let $\mathcal{J}_t \subseteq \{1, \ldots, J\}$ denote the index set of individual univariate predictive regression forecasts selected by the ENet in Equation (2.13). When estimating Equation (2.13), we impose the restriction that $\theta_j \geq 0$ for $j = 1, \ldots, J$. This imposes the economically reasonable requirement that a univariate market excess re-

[5] Again, the results are qualitatively similar in our empirical application in Section 3 if we use the LASSO to estimate Equation (2.11).

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turn forecast be positively related to the realized excess return to be selected by the ENet in Equation (2.13).

Step 3 We compute the C-ENet forecast as

$$\hat{r}_{t+1|t}^{\text{C-ENet}} = \frac{1}{|\mathcal{J}_t|} \sum_{j \in \mathcal{J}_t} \hat{r}_{t+1|t}^{(j)}, \tag{2.14}$$

where $|\mathcal{J}_t|$ is the cardinality of $\mathcal{J}_t$, and $\hat{r}_{t+1|t}^{(j)}$ is given by Equation (2.2) for $j = 1, \ldots, J$.

The usefulness of the C-ENet approach for capturing the relevant information in numerous predictor variables in a manner that guards against overfitting is again ultimately an empirical issue. In our empirical application in Section 3, we find that the C-ENet approach is indeed an effective strategy for forecasting the market excess return.[6]

3 Empirical Application

3.1 Data

We investigate the performance of the strategies discussed in Section 2 for forecasting the monthly S&P 500 excess return. Using data available from Amit Goyal's website,[7] we measure the excess return as the CRSP value-weighted S&P 500 return in excess of the risk-free return (based on the Treasury bill rate).

We consider twelve plausible predictor variables, which are illustrative of popular predictors used by academics and practitioners alike:

Log dividend-price ratio (DP). Log of the twelve-month moving sum of S&P 500 dividends minus the log of the S&P 500 price index.

[6]Observe that all the forecasts that we construct only use data available through t to forecast r_{t+1}, so that the forecasts do not entail "look-ahead" bias.

[7]Aailable at http://www.hec.unil.ch/agoyal/.

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Log earnings-price ratio (*EP*). Log of the twelve-month moving sum of S&P 500 earnings minus the log of the S&P 500 price index.

Volatility (*VOL*). We follow Mele (2007) in measuring the annualized volatility for month t as $\sqrt{\frac{\pi}{2}}\sqrt{12}\hat{\sigma}_t$, where $\hat{\sigma}_t = \frac{1}{12}\sum_{s=1}^{12}|r_{t-(s-1)}|$.

Treasury bill yield (*BILL*). Three-month Treasury bill yield minus the twelve-month moving average of the three-month Treasury bill yield.

Treasury bond yield (*BOND*). Ten-year Treasury bond yield minus the twelve-month moving average of the ten-year Treasury bond yield.

Term spread (*TERM*). Difference in yields on a ten-year Treasury bond and a three-month Treasury bill.

Credit spread (*CREDIT*). Difference in yields on a AAA-rated corporate bond and a ten-year Treasury bond.

Inflation (*PPIG*). Producer price index (PPI) inflation rate.

Industrial production growth (*IPG*). Growth rate of industrial production.

MA(1,12) technical signal [*MA(1,12)*]. An indicator variable that takes a value of one (zero) if the S&P 500 price index is greater than or equal to (less than) the twelve-month moving average of the S&P 500 price index.

MA(3,12) technical signal [*MA(3,12)*]. An indicator variable that takes a value of one (zero) if the three-month moving average of the S&P 500 price index is greater than or equal to (less than) the twelve-month moving average of the S&P 500 price index.

Momentum technical signal [*MOM(6)*]. An indicator variables that takes a value of one (zero) if the S&P 500 price index is greater than or equal to (less than) its value six months ago.

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The data used to construct the predictor variables are from Amit Goyal's website and the Federal Reserve Bank of St. Louis's Federal Reserve Economic Data (FRED).[8] We account for the one-month publication lag in *PPIG* and *IPG*. We follow Neely et al. (2014) in defining indicator variables to include information from technical signals. Figure 1 portrays the twelve predictor variables for the 1927:01 to 2018:12 sample period. Visually, the predictors represent a variety of sources of information.

3.2 Forecasts

We reserve the first two decades of the sample (1927:01 to 1946:12) as the initial in-sample estimation period. This provides us with an adequate number of observations to reasonably reliably estimate the predictive regression coefficients in Equations (2.1) and (2.3). The initial holdout out-of-sample period for computing the C-ENet forecast covers 1947:01 to 1956:12, so that we evaluate the out-of-sample forecasts for 1957:01 to 2018:12. The out-of-sample forecast evaluation period covers more than six decades, which allows us to analyze return predictability under a variety of economic conditions.

Figure 2 depicts the recursive slope coefficient estimates used to compute the predictive regression forecasts in Equations (2.2), (2.4) and (2.10). The black line in each panel delineates the OLS slope coefficient estimates for the multiple predictive regression model in Equation (2.3). The recursive estimates point to problems with the conventional OLS estimates of the multiple predictive regression model slope coefficients—the estimates often have the "wrong" sign (e.g., DP and $BILL$) and reach extreme values, which are manifestations of in-sample overfitting. Overfitting is not surprising when we rely on conventional methods to estimate a relatively high-dimensional predictive regression model in a noisy environment.

The blue line in each panel of Figure 2 delineates the recursive OLS slope coefficient estimates for the univariate predictive regression model in Equation (2.1). Compared to the recursive slope coefficient estimates for the multiple predictive regression model, the

[8] Available at https://fred.stlouisfed.org/.

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univariate estimates are generally much more stable. This reflects the increase in estimation precision afforded by the mitigation of multicollinearity. Of course, the univariate estimates are potentially biased (due to omitted variable bias). However, in light of the bias-efficiency tradeoff, the increase in estimation precision can outweigh the cost of the bias for the purpose of out-of-sample forecasting.

The red line in each panel of Figure 2 shows the ENet slope coefficient estimates for the multiple predictive regression model in Equation (2.3). The shrinkage effect of ENet vis-à-vis OLS estimation of the multiple regression slope coefficients is clear in Figure 2. For some of the predictor variables—such as *BILL*, *IPG*, *MA(3,12)*, and *MOM(6)*—the ENet nearly always shrinks the coefficient estimates all the way to zero. The shrinkage induced by ENet estimation of the multiple predictive regression model in Equation (2.3) should help at least somewhat in alleviating overfitting.

Figure 3 presents the recursive ENet slope coefficient estimates for the Granger and Ramanathan (1984) regression in Equation (2.13) used to compute the C-ENet forecast. Recall that the C-ENet forecast is the average of the individual univariate predictive regression forecasts selected by the ENet in Equation (2.13). Figure 3 indicates that *BOND* is always selected by the ENet for inclusion in the C-ENet forecast, while *DP* is nearly always selected. *VOL*, *TERM*, and *MA(1,12)* are also frequently selected for inclusion in the C-ENet forecast.

Univariate predictive regression forecasts based on Equation (2.2) for the twelve individual predictor variables are shown in Figure 4, while the forecasts based on multiple predictor variables in Equations (2.4), (2.7), (2.10) and (4.7) are presented in Figure 5. The red line in each panel delineates the prevailing mean benchmark forecast. A number of the univariate forecasts in Figure 4 exhibit distinct behavior over the business cycle. In general, the market excess return forecast lies below the prevailing mean benchmark in the months preceding and early months of a recession; the excess return forecast then increases, so that it moves above the prevailing mean benchmark in the later months of and months immediately following a

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recession. The forecasts based on the valuation ratios (DP and EP) also display some longer swings that persist beyond business-cycle frequencies.

Panel A of Figure 5 depicts the OLS multiple predictive regression forecast in Equation (2.4). The forecast generally displays the same type of behavior over the business cycle as many of the univariate predictive regression forecasts in Figure 4. However, the OLS multiple predictive regression forecast is substantially more volatile, and it reaches quite extreme positive and negative values for numerous months over the out-of-sample evaluation period.[9] This is symptomatic of overfitting. Because it maximizes the fit over the estimation sample, conventional estimation is vulnerable to over-responding to noise in the data, especially in a setting with a low signal-to-noise ratio (such as stock return forecasting).

The simple combination forecast in Panel B of Figure 5 is the average of the twelve univariate forecasts in Figure 4. As discussed in Section 2.2, forecast combination exerts a strong shrinkage effect, which is immediately evident from the sharp reduction in forecast volatility as we move from Panel A to Panel B in Figure 5.

ENet estimation of the multiple predictive regression model in Equation (2.3) provides an alternative means for inducing shrinkage in the forecasts. From Panel C of Figure 5, we see that ENet estimation does induce some forecast shrinkage relative to the OLS forecast in Panel A. However, the degree of shrinkage is considerably weaker than that induced by the simple combination forecast in Panel B.

The degree of forecast shrinkage induced by the C-ENet forecast in Panel D of Figure 5 falls between that of the simple combination forecast in Panel B and ENet multiple predictive regression forecast in Panel C. By not necessarily averaging across all of the univariate forecasts, the C-ENet forecast is able to respond more strongly to fluctuations in the univariate forecasts selected by the ENet in the Granger and Ramanathan (1984) regression in Equation (2.13). In this fashion, the C-ENet forecast is designed to mitigate the potential

[9]Note that the forecasts are not annualized in Figures 4 and 5.

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overshrinking induced by the simple combination forecast, while avoiding the complete lack of shrinkage in the OLS multiple predictive regression forecast.

3.3 Statistical Gains

Next, we assess the statistical accuracy of the forecasts in Figures 4 and 5 in terms of mean squared forecast error (MSFE). To this end, we compute the Campbell and Thompson (2008) out-of-sample R^2 statistic:

$$R^2_{\text{OS}} = 1 - \frac{\sum_{s=t_2+1}^{T} \hat{e}_{s|s-1}}{\sum_{s=t_2+1}^{T} \hat{e}^{\text{PM}}_{s|s-1}}, \tag{3.1}$$

where t_2 is the last observation for the initial holdout out-of-sample period, T is the total number of available observations, $\hat{e}_{s|s-1} = r_s - \hat{r}_{s|s-1}$ generically denotes a competing forecast error, and $\hat{e}^{\text{PM}}_{s|s-1} = r_s - \hat{r}^{\text{PM}}_{s|s-1}$ is the prevailing mean forecast error. The R^2_{OS} statistic, which is akin to the familiar in-sample R^2 statistic, measures the proportional reduction in MSFE for a competing forecast vis-à-vis the prevailing mean benchmark forecast. Of course, because monthly stock returns inherently contain a limited predictable component, the R^2_{OS} statistic will be "small."[10] Nevertheless, Campbell and Thompson (2008) argue that a monthly R^2_{OS} statistic of approximately 0.5% indicates an economically significant degree of market excess return predictability.

To gauge whether a competing forecast provides a statistically significant improvement in MSFE relative to the prevailing mean benchmark forecast, we use the Clark and West (2007) adjusted version of the familiar Diebold and Mariano (1995) and West (1996) statistic. The latter is less informative for comparing forecasts from nested models (Clark and McCracken 2001; McCracken 2007). In particular, we use the Clark and West (2007) MSFE-adj statistic to test the null hypothesis that the prevailing mean MSFE is less than or equal to the competing MSFE against the alternative that the competing MSFE is less than the prevailing mean MSFE.

[10] Indeed, if it is "large," it is likely too good to be true!

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Panel A of Table 1 reports R^2_{OS} statistics for the univariate predictive regression forecasts. The R^2_{OS} statistic is negative for the majority of the predictor variables, so that most of the forecasts fail to outperform the prevailing mean benchmark in terms of MSFE. The univariate forecasts based on *VOL*, *BILL*, *BOND*, *TERM*, and *MA(1,12)* produce positive R^2_{OS} statistics, and the MSFE-adj statistics indicate that each of the five forecasts provides a statistically significant reduction in MSFE vis-à-vis the prevailing mean benchmark (at the 10% level of better).[11] Among the positive R^2_{OS} statistics, only that for *BOND* is above the Campbell and Thompson (2008) threshold of 0.5%. Furthermore, we cannot know a priori which of the twelve predictors variables in Panel A will perform the best.

To get a sense of the performance of the individual univariate forecasts over time, Figure 6 shows the cumulative differences in squared forecast errors for the prevailing mean benchmark relative to each univariate forecast (Goyal and Welch 2003, 2008). The cumulative differences in Figure 6 make it straightforward to determine whether a competing forecast is more accurate than the prevailing mean benchmark for any subsample. We simply compare the height of the curve at the start and end of the subsample. If the curve is higher (lower) at the end, then the competing forecast has a lower (higher) MSFE than the prevailing mean benchmark over the subsample. A competing forecast that provides out-of-sample gains on a consistent basis will thus have a predominantly positively sloped curve, while a steeply negatively sloped segment indicates an episode of severe underperformance.

With the exception of *VOL*, the univariate forecasts in Figure 6 fail to provide accuracy gains on a reasonably consistent basis over time. Although there are times when the univariate forecasts substantively outperform the prevailing mean benchmark, there are also periods when they perform much worse than the naive benchmark. Overall, Figure 6 is reminiscent of the findings in Goyal and Welch (2008).

[11] Despite having a negative R^2_{OS} statistic, the MSFE-adj statistic for *DP* is significant. Although this result may seem surprising, it can occur when comparing nested forecasts (Clark and West 2007; McCracken 2007).

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Panel B of Table 1 reports R^2_{OS} statistics for the four forecasts that incorporate information from multiple predictor variables, while Figure 7 depicts the cumulative differences in squared forecast errors for the prevailing mean benchmark relative to the four forecasts. The R^2_{OS} statistic for the OLS multiple predictive regression forecast is -4.33% in Panel B of Table 1, so that the forecast is substantially less accurate than the prevailing mean benchmark over the 1957:01 to 2018:12 evaluation period. Panel A of Figure 7 shows that the forecast exhibits prolonged periods of underperformance relative to the naive benchmark. Overall, the OLS multiple predictive regression forecast apparently suffers from considerable overfitting.

The R^2_{OS} statistic for the simple combination forecast is 1.11% in Panel B of Table 1 (and its MSFE-adj statistic is significant at the 1% level), which is larger than all the R^2_{OS} statistics for the univariate predictive regression forecasts in Panel A. In addition, Panel B of Figure 7 reveals that the simple combination forecast outperforms the prevailing mean benchmark on a reasonably consistent basis over time, as the curve is predominantly positively sloped and displays only limited segments with a negative slope. The strong shrinkage induced by the simple combination forecast is apparently useful for incorporating information from multiple predictor variables in a manner that guards against overfitting.

Although the ENet multiple predictive regression forecast also induces shrinkage, recall from Figure 5 that the degree of shrinkage appears limited. The R^2_{OS} statistic for the ENet forecast is positive but quite close to zero (0.22%) in Panel B of Table 1. Furthermore, Panel C of Figure 7 indicates that the ENet forecast fails to outperform the prevailing mean benchmark consistently over time. Indeed, there are a number of segments where the ENet forecast severely underperforms the prevailing mean. It thus seems that the degree of shrinkage induced by the ENet multiple predictive regression forecast is too weak to effectively guard against overfitting when forecasting the market excess return.

The simple combination forecast exerts a strong—perhaps too strong—shrinkage effect, while the ENet multiple predictive regression forecast induces insufficient shrinkage. Again

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recalling Figure 5, the degree of shrinkage induced by the C-ENet forecast falls between that induced by the simple combination and ENet forecasts. The R^2_{OS} statistic for the C-ENet forecast is a sizable 2.12% in Panel B of Table 1 (and its MSFE-adj statistic is significant at the 1% level). The ENet forecast provides the largest improvement in MSFE among all the competing forecasts relative to the prevailing mean benchmark, and its R^2_{OS} statistic is nearly double that for the simple combination forecast. Panel C of Figure 7 confirms that the C-ENet forecast outperforms the prevailing mean quite consistently over time.[12]

Overall, the C-ENet forecast appears close to a "Goldilocks" shrinkage technique. It induces stronger shrinkage than the ENet forecast, so that it better guards against overfitting. At the same time, it exerts a weaker shrinkage effect than the simple combination forecast, allowing it to incorporate more information from the most relevant univariate forecasts to further improve out-of-sample performance. Recall from Figure 3 that the ENet often selects the univariate predictive regression forecasts based on *DP*, *VOL*, *BOND*, *TERM*, and *MA(1,12)* in the Granger and Ramanathan (1984) regression. This judicious real-time selection of relevant predictor variables to include in the combination forecast demonstrates the value of machine learning for improving out-of-sample market excess return forecasts.

3.4 Economic Gains

In addition to the gains in forecast accuracy documented in Section 3.3, we next show that the C-ENet forecast provides substantial economic gains for an investor in an asset allocation context. Specifically, we consider a mean-variance investor who allocates across risky equities and risk-free Treasury bills each month. The investor's objective function is given by

$$\arg\max_{w_{t+1|t}} w_{t+1|t}\hat{r}_{t+1|t} - 0.5\gamma w^2_{t+1|t}\hat{\sigma}^2_{t+1|t}, \tag{3.2}$$

[12]The out-of-sample gains in Figure 7 are often particularly evident during business-cycle recessions. This is a stylized fact in the literature on market excess return predictability (e.g., Rapach et al. 2010; Henkel et al. 2011; Rapach and Zhou 2013).

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where γ is the coefficient of relative risk aversion, $w_{t+1|t}$ $(1-w_{t+1|t})$ is the investor's allocation to equities (risk-free bills) in month $t+1$, and $\hat{r}_{t+1|t}$ $(\hat{\sigma}^2_{t+1|t})$ is the market excess return point (variance) forecast used by the investor. The well-known solution to Equation (3.2) takes the form:[13]

$$w^*_{t+1|t} = \left(\frac{1}{\gamma}\right)\left(\frac{\hat{r}_{t+1|t}}{\hat{\sigma}^2_{t+1|t}}\right). \tag{3.3}$$

To measure the economic value of return predictability to the investor, we first analyze portfolio performance when the investor relies on the prevailing mean forecast—which ignores return predictability—to determine the optimal equity allocation in Equation (3.3). We then analyze portfolio performance when the investor instead uses the C-ENet forecast to select the optimal allocation. In both cases, the investor uses the sample variance computed over a 60-month rolling window to forecast the variance. We assume that $\gamma = 5$.[14]

When the investor relies on the prevailing mean benchmark forecast, the optimal portfolio earns an annualized average excess return of 4.30% for the 1957:01 to 2018:12 forecast evaluation period. Together with an annualized volatility of 12.87%, the portfolio's annualized Sharpe ratio is 0.33. If the investor uses the C-ENet forecast in lieu of the prevailing mean in Equation (3.3), then the optimal portfolio yields an annualized average excess return of 6.29% and volatility of 9.76%. These translate into a substantive annualized Sharpe ratio of 0.64, which is nearly twice as large as that for the portfolio based on the prevailing mean.[15] Moreover, the gain in certainty equivalent return (CER) indicates that the investor would be willing to pay a hefty annualized management fee of 375 bps to switch from the prevailing mean to the C-ENet forecast.

The red and black lines in Panel A of Figure 8 depict the equity allocations for the optimal portfolios based on the prevailing mean and C-ENet forecasts, respectively. There

[13]To prevent what could be construed as impractical allocations, we impose the restriction that $-1 \leq w_{t+1|t} \leq 2$. As shown in Panel A of Figure 8, the constraint is rarely binding.

[14]The results are qualitatively similar for reasonable alternative value for γ.

[15]The annualized Sharpe ratio for the market excess return for 1957:01 to 2018:12 is 0.42, so that the optimal portfolio based on the C-ENet forecast delivers a Sharpe ratio that it is over 50% larger than that for the market portfolio.

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are numerous months when the C-ENet forecast leads to a markedly different allocation than the prevailing mean benchmark. Such reallocations appear quite valuable to the investor, as they substantially improve portfolio performance. This is further illustrated in Panel B of Figure 8, which shows the log cumulative excess returns for the two portfolios. For example, the portfolio based on the C-ENet forecast typically suffers smaller drawdowns than the portfolio based on the prevailing mean benchmark; indeed, the maximum drawdown for the former is only half as large as that for the latter (29% and 58%, respectively).

The optimal portfolio based on the C-ENet forecast also generates a higher Sharpe ratio and annualized CER gain than the optimal portfolios based on any of the twelve univariate predictive regression forecasts, as well as the OLS and ENet multiple predictive regression and simple combination forecasts.[16] In sum, the C-ENet forecast delivers the best overall performance in terms of statistical accuracy and investor value.

4 Cross-Sectional Return Forecasts

In this section, we outline the cross-sectional return forecasting procedures in Han et al. (2019), which parallel the time-series strategies in Section 2.[17] As Lewellen (2015) argues, the appropriate target is the cross-sectional dispersion in firm returns (and not the cross-sectional average return per se); we can simply adjust all of the cross-sectional return forecasts up or down as needed to reflect our forecast of the overall market return.

Consider the following cross-sectional multiple regression model for month t:

$$r_{i,t} = \alpha_t + \sum_{j=1}^{J} \beta_{j,t} z_{i,j,t-1} + \varepsilon_{i,t}, \tag{4.1}$$

for $i = 1, \ldots, I_t$, where $z_{i,j,t}$ is the jth characteristic for firm i in month t, and I_t is the number of firm observations available in month t. Analogously to Equation (2.4), a conventional

[16] The complete results are available upon request from the authors.

[17] See Han et al. (2019) for further details on the construction of the cross-sectional stock return forecasts discussed in this section.

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cross-sectional return forecast based on Equation (4.1) is given by

$$\hat{r}^{\text{OLS}}_{i,t+1|t} = \hat{\alpha}^{\text{OLS}}_t + \sum_{j=1}^{J} \hat{\beta}^{\text{OLS}}_{j,t} z_{i,j,t}, \tag{4.2}$$

where $\hat{\alpha}^{\text{OLS}}_t$ and $\hat{\beta}^{\text{OLS}}_{j,t}$ are the OLS estimates of α_t and $\beta_{j,t}$, respectively, for $j = 1, \ldots, J$ in Equation (4.1). The literature on cross-sectional returns has investigated many characteristics (e.g., Harvey et al. 2016), so that the number of plausible predictors J to include in Equation (4.1) is quite large. As in the time-series context, conventional estimation of Equation (4.1) is prone to overfitting.

It is common to smooth the OLS coefficient estimates in Equation (4.2) over time (e.g., Lewellen 2015; Green et al. 2017). However, this appears inadequate for mitigating overfitting. To see this, we use data for 84 firm characteristics starting in 1980:01 to forecast cross-sectional stock returns.[18] The firm characteristics are similar to those used in Lewellen (2015), Green et al. (2017), and Freyberger et al. (forthcoming). We compute out-of-sample cross-sectional stock returns for 1990:01 to 2018:06 using Equation (4.2), although we smooth the OLS coefficient estimates over time using an expanding window before forming the forecasts.

Lewellen (2015) provides an informative predictive slope to assess the ability of a forecast to track cross-sectional expected returns. We compute the predictive slope in two steps. In the first step, we estimate a cross-sectional version of a Mincer and Zarnowitz (1969) regression for month t:

$$r_{i,t} = \phi_t + \psi_t \hat{r}_{i,t+1|t} + \varepsilon_{i,t}, \tag{4.3}$$

where $\hat{r}_{i,t+1|t}$ generically denotes a cross-sectional return forecast. We estimate ψ_t via OLS for each month over the forecast evaluation period. In the second step, we compute the time-series average of the monthly cross-sectional slope coefficient estimates in Equation (4.3); we

[18] To minimize the effects of micro-cap stocks, we omit stocks with market capitalization below the NYSE median.

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denote the average predictive slope estimate by $\hat{\psi}$. As discussed by Lewellen (2015), $\psi = 1$ indicates that the forecasts are unbiased with respect to the cross-sectional dispersion in expected returns: a percentage point increase in the forecast corresponds to a percentage point increase in the realized return on average. If $\psi < 1$, then the cross-sectional forecasts are characterized by overfitting, because a percentage point increase in the forecast corresponds to a less than percentage point increase in the realize return on average. Alternatively, $\psi > 1$ signals that the forecasts are conservative, in that a percentage point increase in the forecast coincides with a more than percentage point increase in the actual return on average.

For the OLS multiple regression forecast, the $\hat{\psi}$ estimate is 0.10, while the average cross-sectional R^2 statistic in Equation (4.3) is 0.61%. Based on its standard error, the $\hat{\psi}$ estimate is significantly greater than zero. However, it is also significantly below unity, so that the conventional OLS forecast exhibits significant cross-sectional overfitting, as anticipated.

Analogously to Equation (2.7), Han et al. (2019) propose a simple combination strategy for forecasting cross-sectional returns:

$$\hat{r}^{\mathrm{C}}_{i,t+1|t} = \frac{1}{J}\sum_{j=1}^{J} \hat{r}^{(j)}_{i,t+1|t}, \tag{4.4}$$

where

$$\hat{r}^{(j)}_{i,t+1|t} = \hat{\alpha}^{(j)}_t + \hat{\beta}^{(j)}_t z_{i,j,t}, \tag{4.5}$$

and $\hat{\alpha}^{(j)}_t$ and $\hat{\beta}^{(j)}_t$ are the OLS estimates of the intercept and slope coefficients, respectively, in the following cross-sectional univariate regression model:

$$r_{i,t} = \alpha_t + \beta_t z_{i,j,t-1} + \varepsilon_{i,t}, \tag{4.6}$$

for $j = 1, \ldots, J$. Again like the time-series case, the simple combination forecast in Equation (4.4) exerts a strong shrinkage effect.[19] The $\hat{\psi}$ estimate is 2.27 for the combination

[19]Observe that we do not smooth the OLS coefficient estimates over time in Equation (4.5) when forming the cross-sectional forecast, as the simple combination forecast already induces strong shrinkage.

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forecast (with an average cross-sectional R^2 of 4.85%), which is significantly greater than both zero and unity. The simple combination approach thus appears to overshrink the forecast, rendering it overly conservative on average.

Han et al. (2019) recommend refining the simple combination forecast in Equation (4.4) using machine learning techniques. Specifically, they suggest selecting the individual characteristics to include in the combination forecast by using the LASSO or ENet to estimate a cross-sectional Granger and Ramanathan (1984) regression that relates month-t realized returns to the univariate forecasts in Equation (4.5). The cross-sectional C-ENet forecast is given by

$$\hat{r}_{i,t+1|t}^{\text{C-ENet}} = \frac{1}{|\mathcal{J}_t|} \sum_{j \in \mathcal{J}_t} \hat{r}_{i,t+1|t}^{(j)}, \tag{4.7}$$

where $\mathcal{J}_t \subseteq \{1, \ldots, J\}$ is the index set of cross-sectional univariate forecasts selected by the ENet in the month-t Granger and Ramanathan (1984) regression. As in the time-series case, the refinement proves efficacious: the $\hat{\psi}$ estimate for the C-ENet forecast is 1.25 (with an average cross-sectional R^2 statistic of 3.65%), which is significantly greater than zero but insignificantly different from unity.

Han et al. (2019) also develop a cross-sectional forecast that blends the conventional OLS multiple regression and C-ENet forecasts using the notion of forecast encompassing (Harvey et al. 1998). The $\hat{\psi}$ estimate for their encompassing C-ENet forecast is 1.10 (with an average cross-sectional R^2 statistic of 2.69%), which is even closer to unity. The $\hat{\psi}$ estimate is significantly greater than zero and insignificantly different from unity. In an extensive empirical application involving more than 100 firm characteristics, Han et al. (2019) show that their encompassing C-ENet approach provides the most accurate forecasts to date of the cross-sectional dispersion in expected stock returns.

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5 Conclusion

Extending the cross-sectional return forecasting procedures developed by Han et al. (2019), this paper introduces some new machine learning methods for time-series stock return forecasting. Our empirical application focuses on forecasting the US market excess return, a central issues in finance. Despite evidence of in-sample predictability, Goyal and Welch (2008) show that conventional forecasts based on many popular predictor variables from the literature fail to provide out-of-sample gains on a consistent basis over time. Using simple forecast combination, Rapach et al. (2010) are seemingly the first to provide evidence of consistent out-of-sample market excess return predictability. The methods proposed in this paper use the ENet to refine the simple combination forecast, and we find that the ENet refinement embodied in our C-ENet forecast indeed generates substantive further improvements in out-of-sample market excess return predictability.

Machine learning techniques are often criticizes for being "black boxes." However, by performing variable selection, the ENet (and LASSO) is a machine learning technique that facilitates economic interpretation. In our empirical application, our C-ENet approach consistently identifies the dividend-price ratio, volatility, Treasury bond yield, term spread, and a popular technical signal as relevant market excess return predictors. The identification of the most relevant out-of-sample market excess return predictors from among the plethora of predictors from the literature provides a useful guide for researchers in constructing theoretical asset pricng models. As analyzed by Han et al. (2019) in a cross-sectional context, the C-ENet approach can also be used to identify the most relevant firm characteristics for explaining cross-sectional expected returns. More generally, since countless questions in asset pricing and corporate finance are related to either time-series or cross-sectional return forecasting, the C-ENet approach discussed in this paper should provide a valuable resource for researchers and practitioners alike.

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Table 1: Forecast accuracy

(1)	(2)	(3)	(4)	(5)	(6)
Forecast	R^2_{OS}	MSFE-adj	Forecast	R^2_{OS}	MSFE-adj
Panel A: Individual predictor variables					
DP	−0.40%	1.81**	*CREDIT*	−0.15%	−0.16
EP	−1.47%	0.79	*PPIG*	−0.50%	0.18
VOL	0.42%	2.58***	*IPG*	−0.02%	−0.04
BILL	0.15%	1.63*	*MA(1,12)*	0.28%	1.38*
BOND	1.04%	3.37***	*MA(3,12)*	−0.15%	0.32
TERM	0.26%	1.59*	*MOM(6)*	−0.04%	0.71
Panel B: Multiple predictor variables					
OLS multiple predictive regression	−4.33%	2.73***	ENet multiple predictive regression	0.22%	3.11***
Simple combination	1.11%	3.70***	C-ENet	2.12%	4.05***

The table reports out-of-sample R^2 (R^2_{OS}) statistics for monthly market excess return forecasts for 1957:01 to 2018:12. The out-of-sample R^2 statistic is the percent reduction in mean squared forecast error (MSFE) for a competing forecast vis-á-vis the prevailing mean benchmark forecast. The competing forecasts in Panel A are based on univariate predictive regression models estimated via ordinary least squares. The predictor variable definitions in Panel A are provided in Section 3.1. The competing forecasts in Panel B use all twelve predictor variables. The OLS (ENet) multiple predictive regression forecast is based on a multiple predictive regression model with all twelve predictor variables estimated via ordinary least squares (the elastic net). The simple combination forecast is the average of the individual univariate predictive regression forecasts in Panel A. The C-ENet forecast is an average of the individual univariate predictive regression forecasts in Panel A selected by the elastic net in a Granger and Ramanathan (1984) regression. MSFE-adj is the Clark and West (2007) statistic for testing the null hypothesis that the benchmark MSFE is less than or equal to the competing MSFE against the alternative hypothesis that the benchmark MSFE is greater than the competing MSFE; *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

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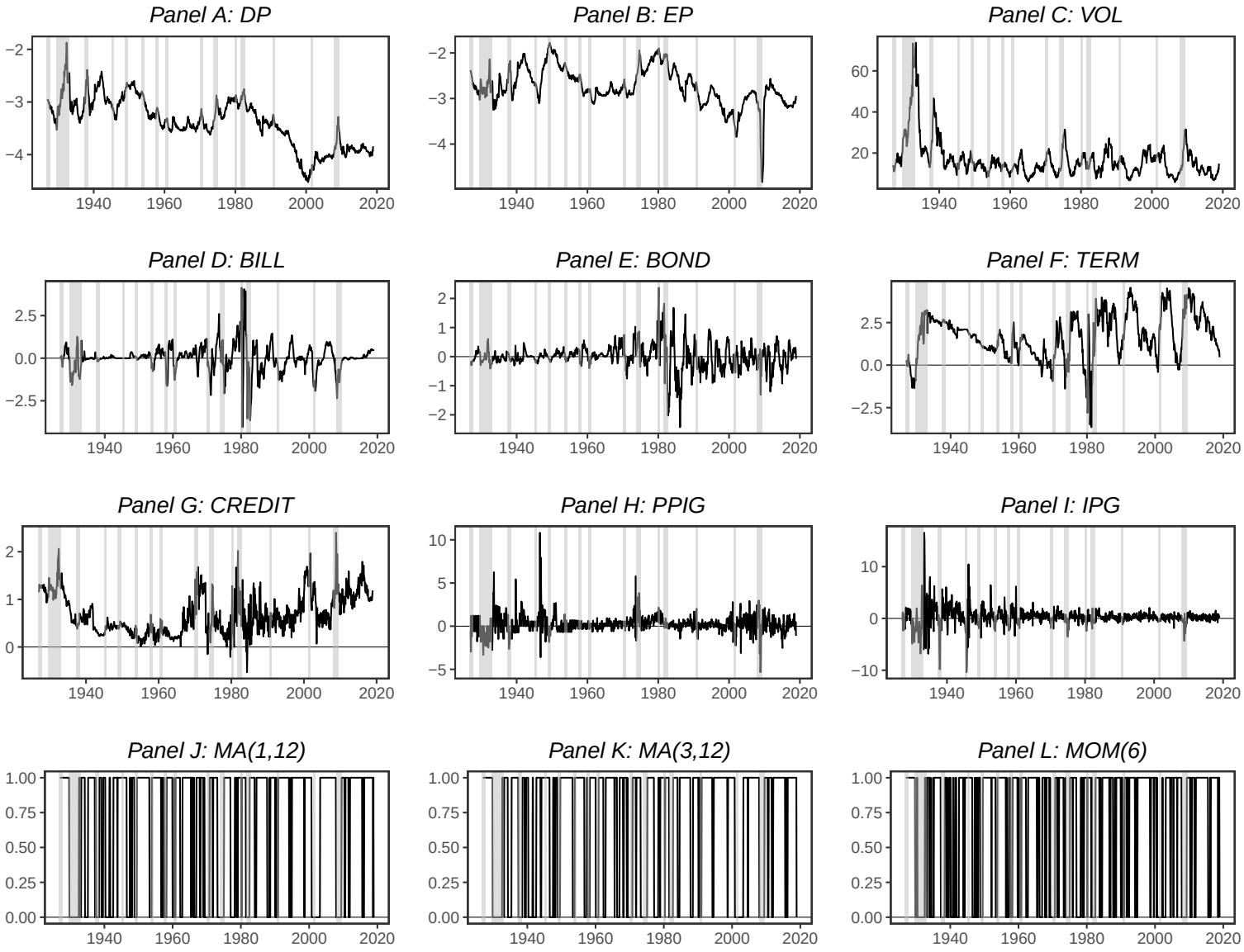


Figure 1: Predictor variables

The figure depicts twelve predictor variables for 1927:01 to 2018:12. The predictor variable definitions are provided in Section 3.1. Vertical bars delineate business-cycle recessions as dated by the NBER.

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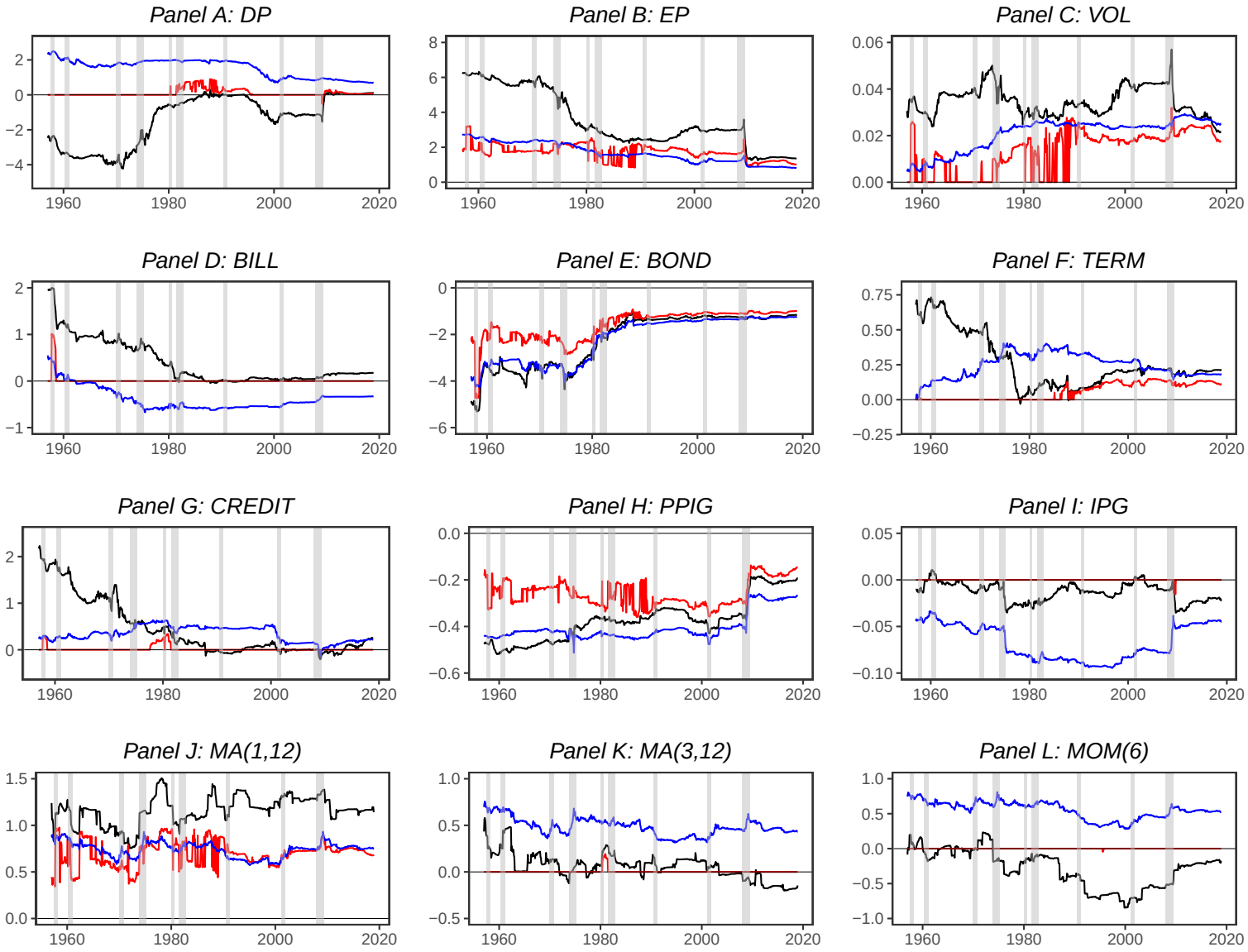


Figure 2: Recursive predictive regression model slope coefficient estimates

The figure depicts recursive predictive regression model slope coefficient estimates used to compute market excess return forecasts. The black (red) lines delineate multiple predictive regression model slope coefficients estimated via ordinary least squares (the elastic net) for the predictor variable in the panel heading. The blue lines delineate univariate predictive regression model slope coefficients estimated via ordinary least squares. The predictor variable definitions are provided in Section 3.1. Vertical bars delineate business-cycle recessions as dated by the NBER.

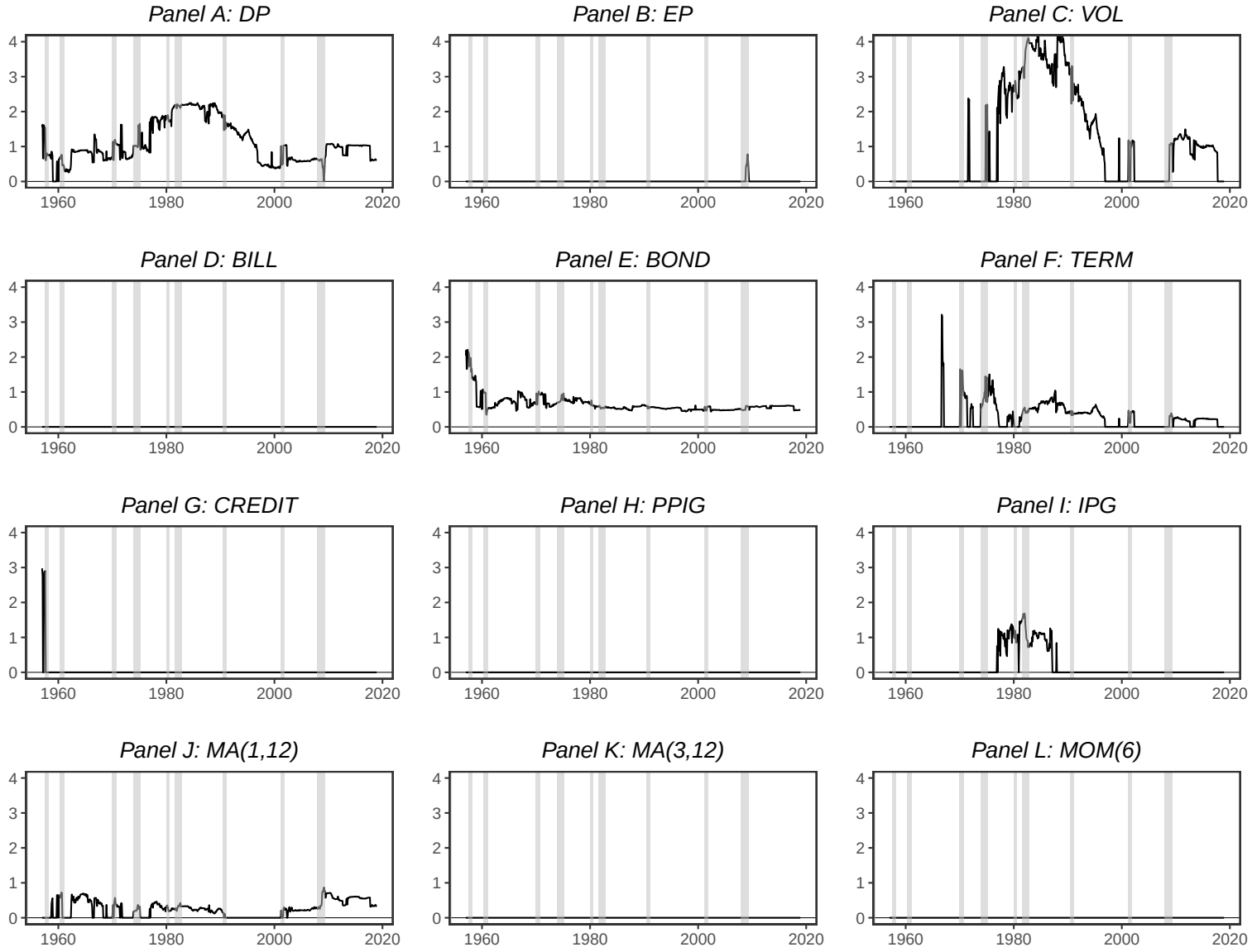


Figure 3: Recursive elastic net Granger and Ramanathan (1984) regression slope coefficient estimates

The figure depicts recursive elastic net slope coefficient estimates for a Granger and Ramanathan (1984) regression based on univariate predictive regression forecasts for twelve individual predictor variables. The predictor variable definitions are provided in Section 3.1. Vertical bars delineate business-cycle recessions as dated by the NBER.

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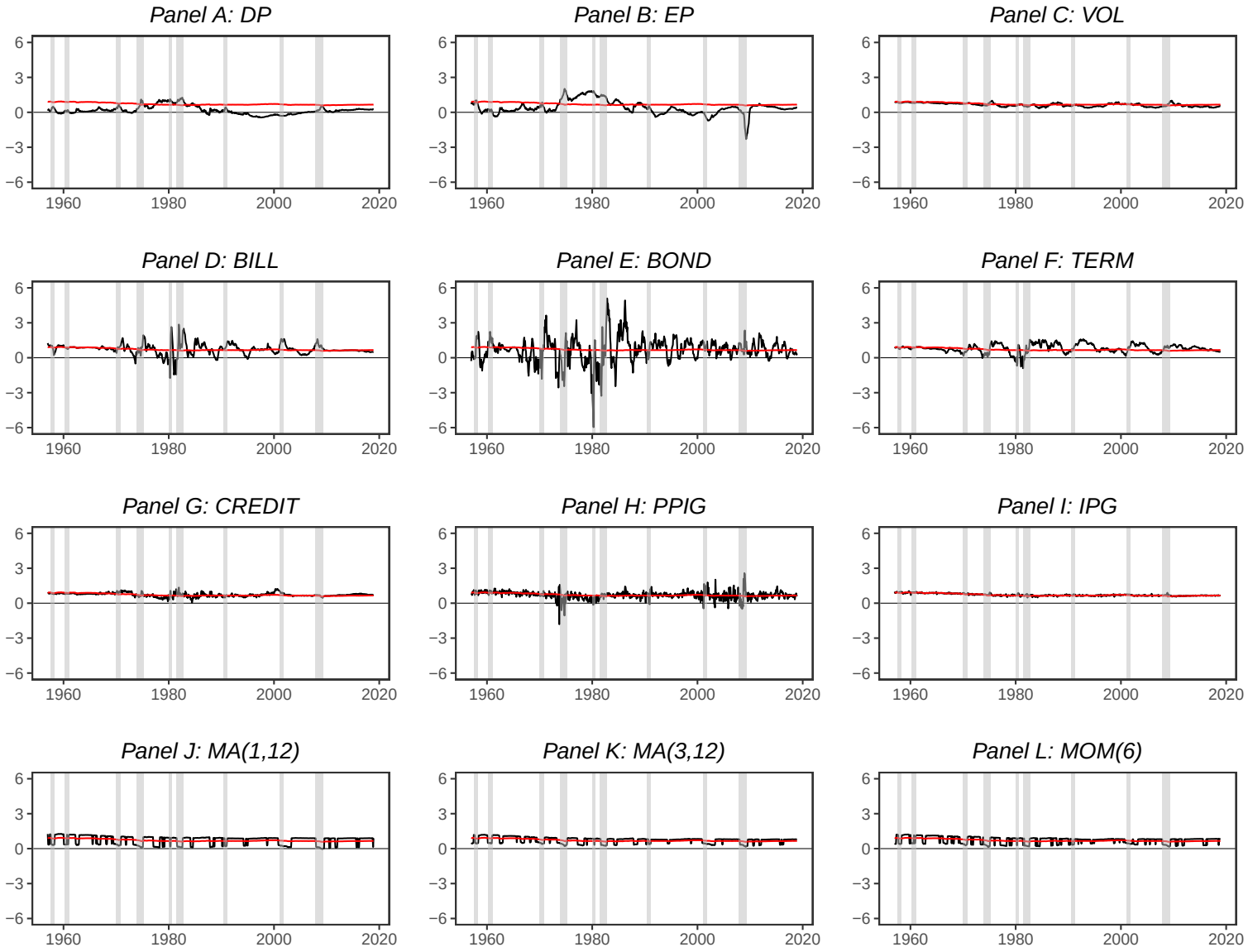


Figure 4: Market excess return forecasts based on individual predictor variables

The figure depicts out-of-sample market excess return forecasts (in percent) for 1957:01 to 2018:12. The black line in each panel delineates a forecast based on ordinary least squares estimation of a univariate predictive regression model with the predictor variable in the panel heading. The predictor variable definitions are provided in Section 3.1. The red line in each panel delineates the prevailing mean benchmark forecast. Vertical bars delineate business-cycle recessions as dated by the NBER.

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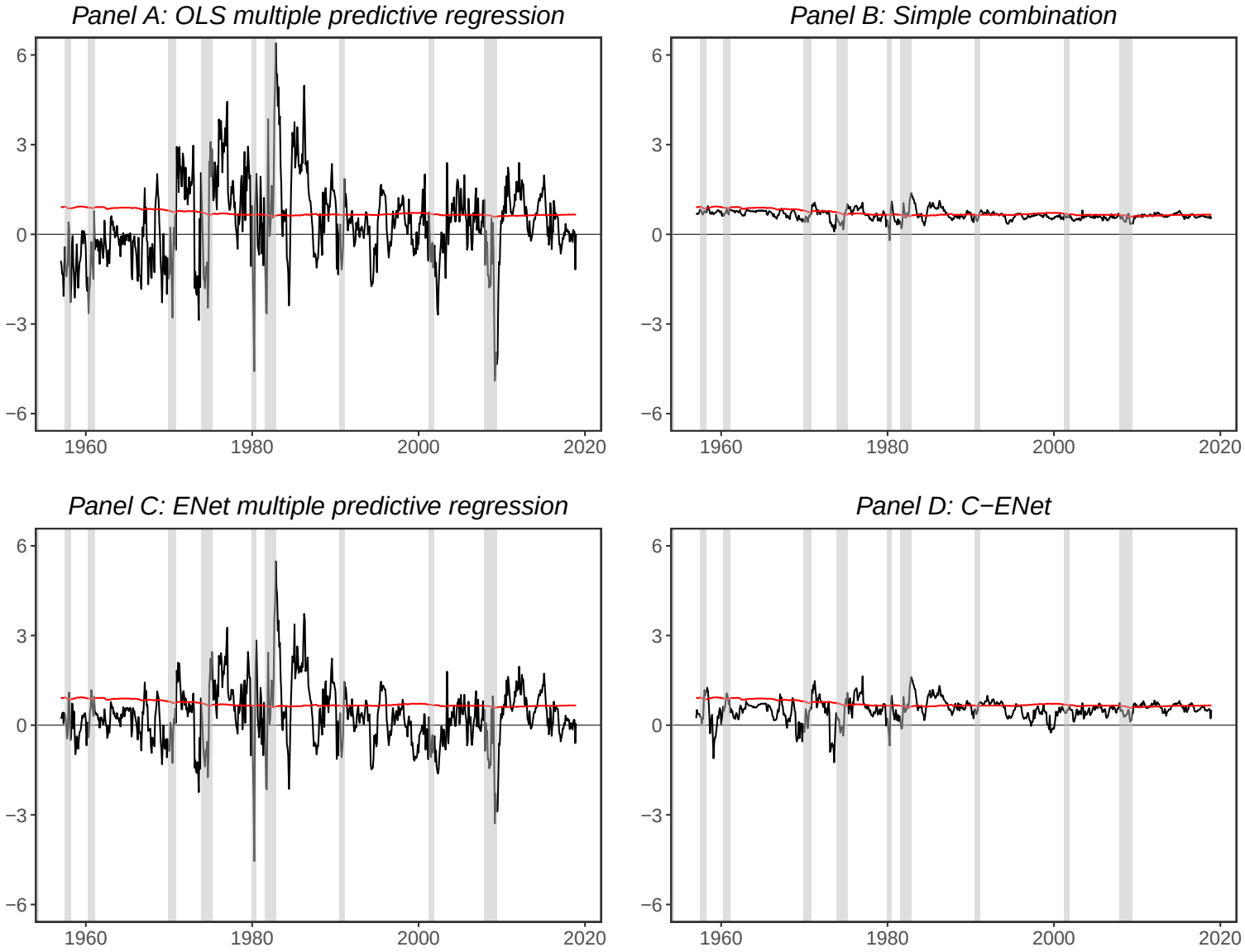


Figure 5: Market excess return forecasts based on multiple predictor variables

The figure depicts out-of-sample market excess return forecasts (in percent) for 1957:01 to 2018:12 based on twelve predictor variables using the method in the panel heading. ENet stands for elastic net; C-ENet is the combination elastic net forecast. Vertical bars delineate business-cycle recessions as dated by the NBER.

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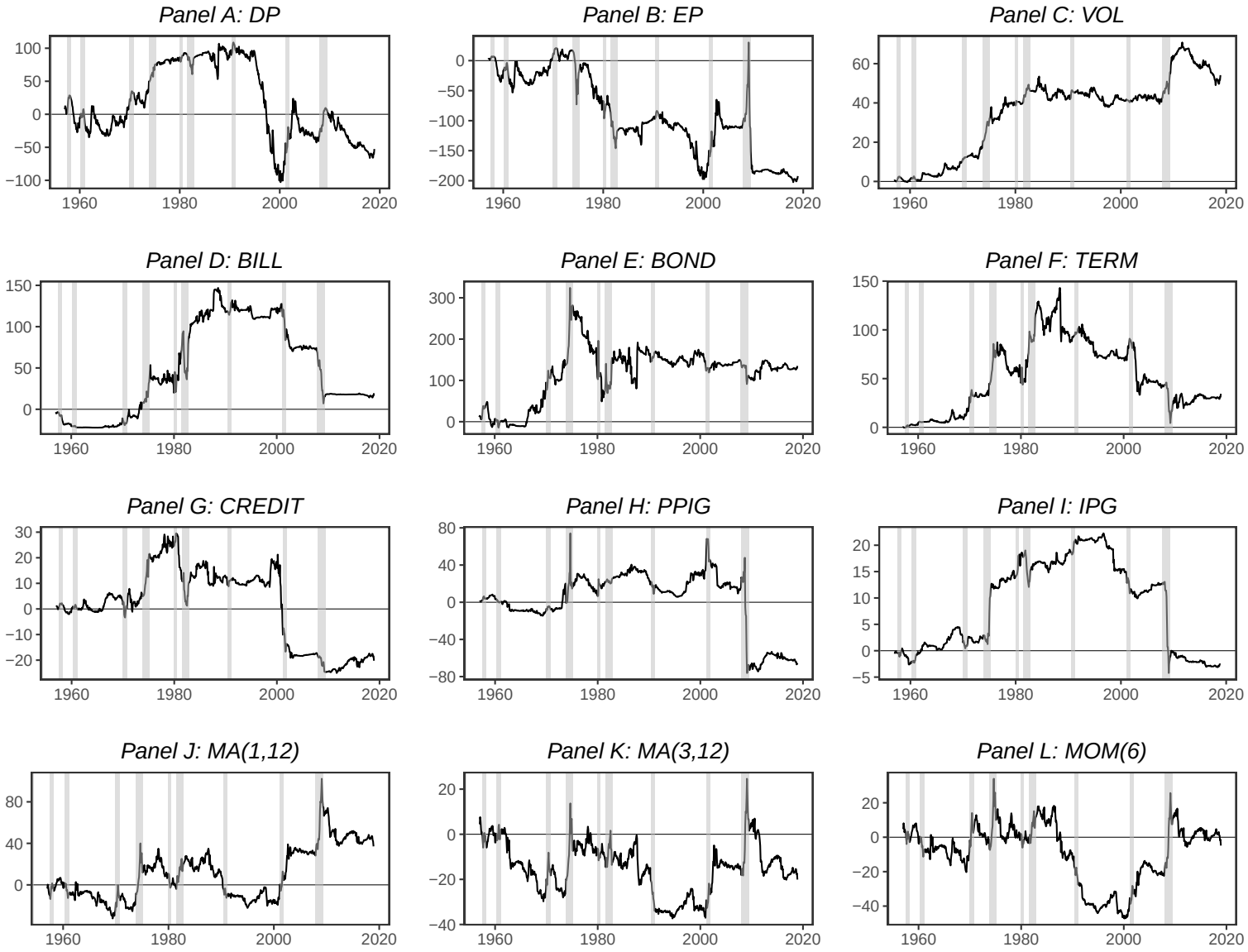


Figure 6: Cumulative differences in squared forecast errors for market excess return forecasts based on individual predictor variables

The figure depicts cumulative differences in squared forecast errors between benchmark and competing out-of-sample market excess return forecasts for 1957:01 to 2018:12. The competing forecast is based on ordinary least squares estimation of a univariate predictive regression model with the predictor variable in the panel heading; the benchmark forecast is the prevailing mean. The predictor variable definitions are provided in Section 3.1. Vertical bars delineate business-cycle recessions as dated by the NBER.

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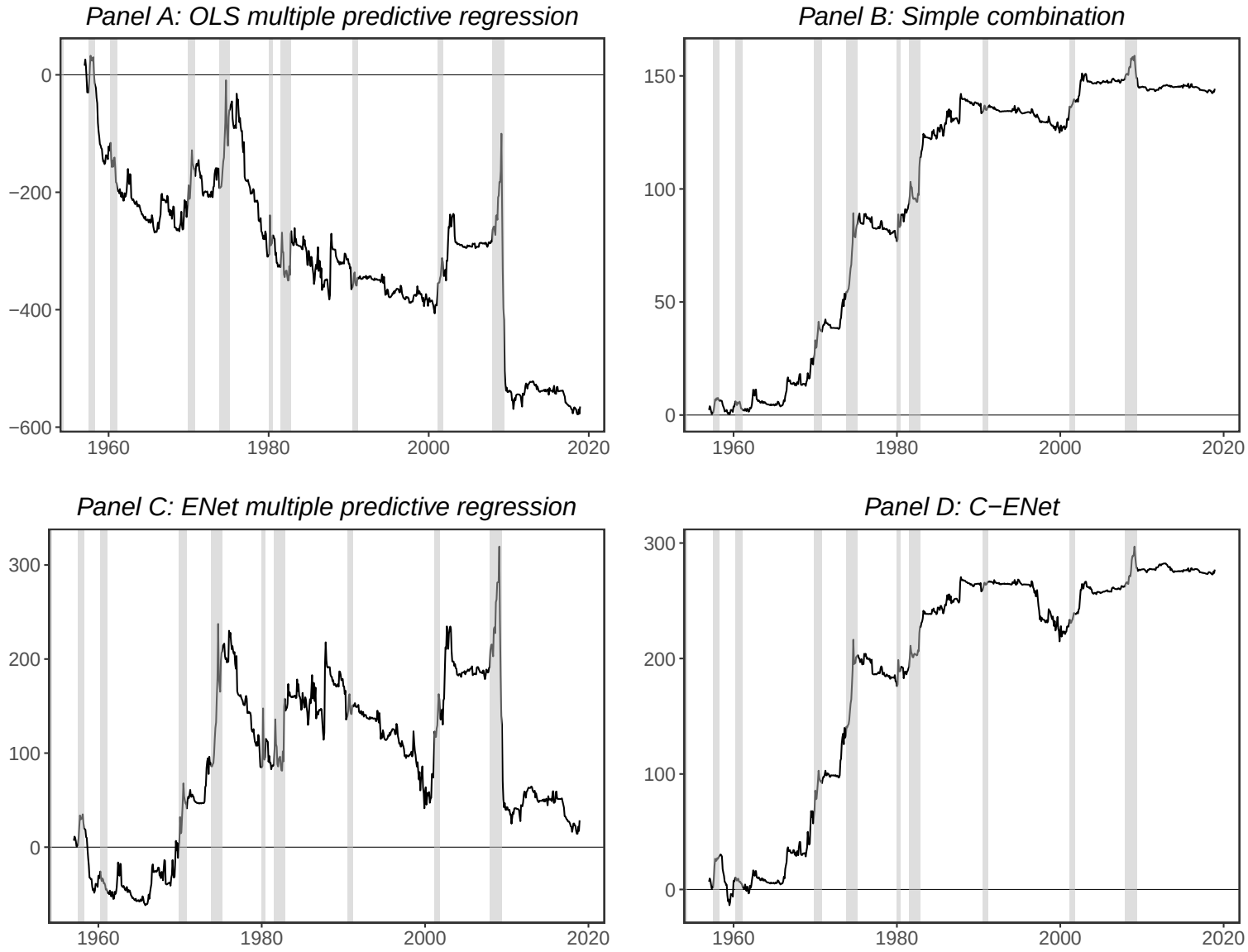


Figure 7: Cumulative differences in squared forecast errors for market excess return forecasts based on multiple predictor variables

The figure depicts cumulative differences in squared forecast errors between benchmark and competing out-of-sample market excess return forecasts for 1957:01 to 2018:12. The competing forecast is based on twelve predictor variables using the method in the panel heading; the benchmark forecast is the prevailing mean. ENet stands for elastic net; C-ENet is the combination elastic net forecast. Vertical bars delineate business-cycle recessions as dated by the NBER.

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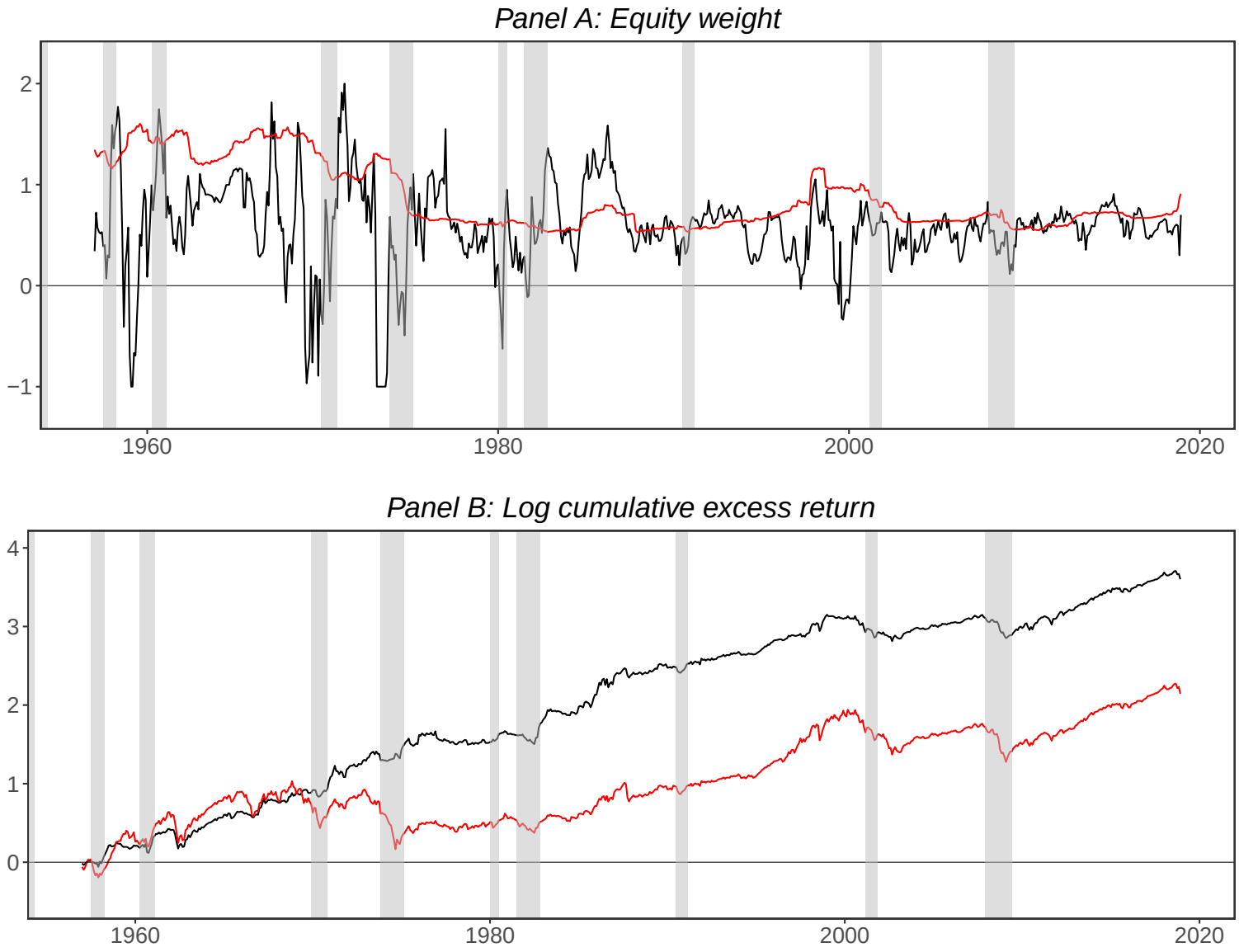


Figure 8: Equity allocations and cumulative excess returns.

The black (red) line in Panel A delineates the equity weight for a mean-variance investor with a coefficient of relative risk aversion of five who uses the C-ENet (prevailing mean) forecast when allocating between equities and risk-free Treasury bills for 1957:01 to 2018:12. The lines in Panel B show the corresponding log cumulative excess for the two portfolios. Vertical bars delineate business-cycle recessions as dated by the NBER.